

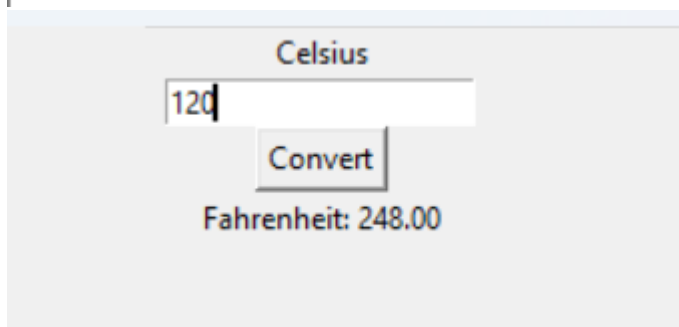
```
#Task 1
import tkinter as tk

def convert():
    c = float(entry.get())
    f = (c * 9/5) + 32
    result.config(text=f"Fahrenheit: {f:.2f}")

root = tk.Tk()
tk.Label(root, text="Celsius").pack()
entry = tk.Entry(root)
entry.pack()

tk.Button(root, text="Convert", command=convert).pack()
result = tk.Label(root, text="")
result.pack()

root.mainloop()
```



```

#Task 2

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tkinter as tk

def generate():
    nums = np.random.randint(1,100,20)
    df = pd.DataFrame(nums, columns=["Numbers"])

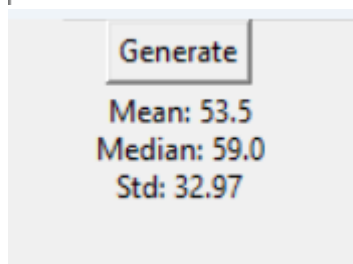
    mean = np.mean(nums)
    median = np.median(nums)
    std = np.std(nums)

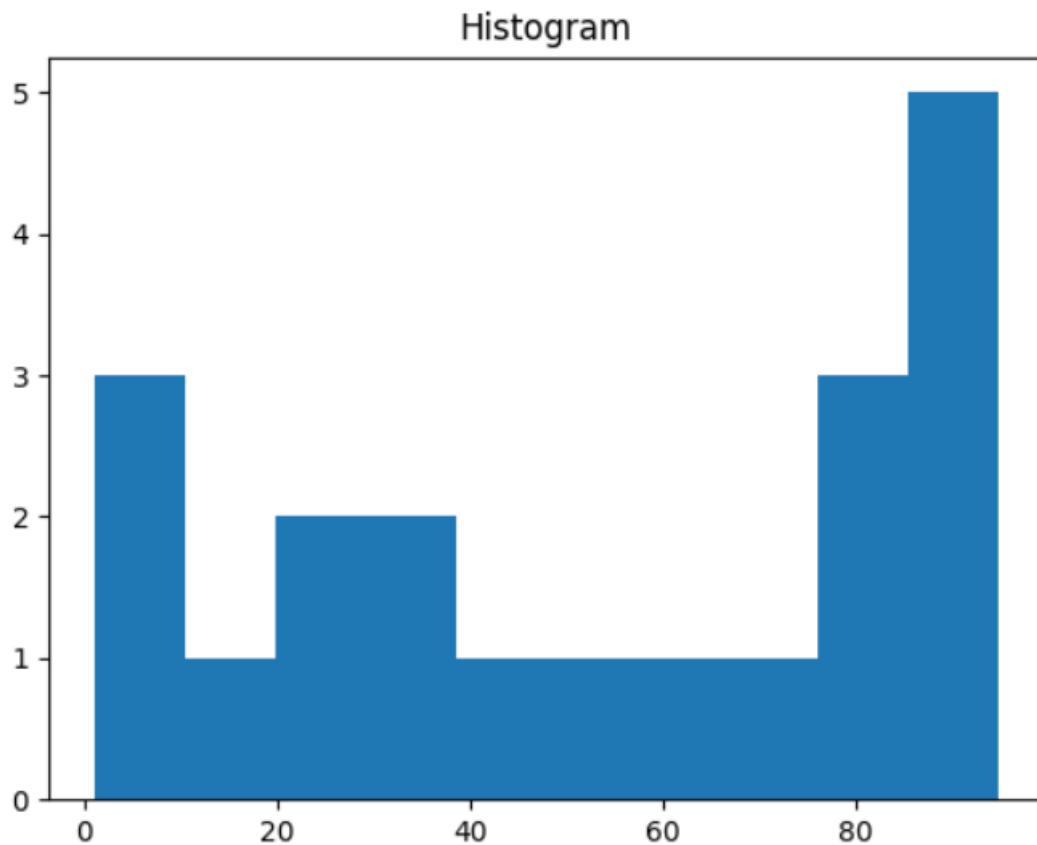
    result.config(text=f"Mean: {mean}\nMedian: {median}\nStd: {std:.2f}")

    plt.hist(nums)
    plt.title("Histogram")
    plt.show()

root = tk.Tk()
tk.Button(root, text="Generate", command=generate).pack()
result = tk.Label(root, text="")
result.pack()
root.mainloop()

```





#Task 3

```
class Book:
    def __init__(self, title, author, price):
        self.title = title
        self.author = author
        self.price = price

    def display(self):
        print(self.title, self.author, self.price)
```

```
b1 = Book("Python", "Guido", 500)
b1.display()
```

```
===== RESTART: C:/User
Python Guido 500
```

#Task 4

```
class Product:
    def __init__(self, name, price):
        self.name = name
        self.price = price

class Cart:
    def __init__(self):
        self.items = []

    def add(self, product):
        self.items.append(product)

    def total(self):
        return sum(p.price for p in self.items)

p1 = Product("Phone", 10000)
p2 = Product("Laptop", 50000)

cart = Cart()
cart.add(p1)
cart.add(p2)

print("Total:", cart.total())
```

```
===== RESTART:
Total: 60000
> |
```

#Task 5

```
class ATM:
    def __init__(self, balance=0):
        self.balance = balance

    def check_balance(self):
        print("Balance:", self.balance)

    def deposit(self, amt):
        self.balance += amt

    def withdraw(self, amt):
        if amt <= self.balance:
            self.balance -= amt
        else:
            print("Insufficient balance")

atm = ATM(1000)
atm.deposit(500)
atm.withdraw(300)
atm.check_balance()
```

```
===== RESTART: C:/\
Balance: 1200
```

.>

#Task 6

```
class Student:
    def __init__(self, name):
        self.name = name

class Teacher:
    def __init__(self, name):
        self.name = name

class Course:
    def __init__(self, name):
        self.name = name
        self.students = []
        self.teacher = None

    def add_student(self, student):
        self.students.append(student)

    def assign_teacher(self, teacher):
        self.teacher = teacher

    def display(self):
        print("Course:", self.name)
        print("Teacher:", self.teacher.name)
        print("Students:", [s.name for s in self.students])

c = Course("AIML")
t = Teacher("Rakesh")
s1 = Student("Subrahmanya")

c.assign_teacher(t)
c.add_student(s1)
c.display()
```

===== RESTART: C:/Users/ki

```
Course: AIML
Teacher: Rakesh
Students: ['Subrahmanya']
```

#Task 7

```
for num in range(1,101):  
    if num > 1:  
        for i in range(2,num):  
            if num % i == 0:  
                break  
        else:  
            print(num)
```

===== RESTART: C:/

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67
71
73
79
83
89
97

```
#Task 8
```

```
s = "ABCDEFGH IJKL"
```

```
print(s[2:9:2])
```

```
print(s[:3:-1])
```

```
print(s[::-1][:-1])
```

```
print(s[10:1:-3])
```

```
print(s[:4])
```

```
s = "Python String Slicing Example"
```

```
print(s[13::-1])
```

```
print(s[7:])
```

```
print(s[:3])
```

```
print(s[:4])
```

```
print(s[:-8:-1])
```

```
print(s[:5])
```

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===== RESTART: C:/Us
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String Slicing Example
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```

#Task 9

List 1

```
lst = ["Cat", "Dog", "Lion", "Tiger", "Rabbit", "Monkey"]
```

```
print([lst[2]])
print([lst[5], lst[4]])
print(lst[3:0:-1])
print([lst[0], lst[3]])
print([lst[3], lst[0]])
print([lst[5], lst[2]])
print([lst[4], lst[2], lst[0]])
print(lst[::-1])
```

Modify list

```
lst2 = ["apple", "banana", "cherry"]
lst2.append("orange")
lst2.insert(1, "mango")
lst2.extend(["kiwi", "grape"])
print(lst2)
```

Number list

```
lst3 = [10, 20, 30, 40, 50]
lst3[2] = 300
lst3[1] = 200
print(lst3)
```

```
lst4 = [1, 2, 3]
lst4.insert(1, 100)
lst4.append(999)
print(lst4)
```

```
lst5 = [10, 20, 30, 40, 50]
lst5.append(60)
lst5.insert(0, 5)
lst5.extend([70, 80, 90])
print(lst5)
```

```
lst6 = [42, 3.14, "Hello", True]
lst6[0] = 2.718
lst6.append(1000)
lst6.insert(1, False)
lst6[0] = 5
print(lst6)
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 3
['Lion']
['Monkey', 'Rabbit']
['Tiger', 'Lion', 'Dog']
['Cat', 'Tiger']
['Tiger', 'Cat']
['Monkey', 'Lion']
['Rabbit', 'Lion', 'Cat']
['Monkey', 'Rabbit', 'Tiger', 'Lion', 'Dog', 'Cat']
['apple', 'mango', 'banana', 'cherry', 'orange', 'kiwi', 'grape']
[10, 200, 300, 40, 50]
[1, 100, 2, 3, 999]
[5, 10, 20, 30, 40, 50, 60, 70, 80, 90]
[5, False, 3.14, 'Hello', True, 1000]
```

#Task 10

```
s = "Hello World, Welcome to Python!"
```

```
print(s.upper())
print(s.lower())
print(s.split(" "))
print(s.split("o"))
print(s.replace("W", "X"))
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/i
HELLO WORLD, WELCOME TO PYTHON!
hello world, welcome to python!
['Hello', 'World,', 'Welcome', 'to', 'Python!']
['Hell', ' W', 'rld, Welc', 'me t', ' Pyth', 'n!']
Hello Xorld, Xelcome to Python!
```